V.L Memorial Public School

475 Vijay Nagar

CHAPTER-11-QUESTIONS-CLASS7TH-MATHS

Class 07 - Maths (Ganita Prakash)

Section A

1. What is the H.C.F of 40, 56 and 60? **[1]**

   a) 16 b) 8

   c) 6 d) 4

2. What is the L.C.M of 21, 35, 40? **[1]**

   a) 840 b) 700

   c) 800 d) 740

3. Which number will divide both 306 and 36 exactly? **[1]**

   a) 7 b) 12

   c) 24 d) 18

4. LCM of two numbers is 180. Then which of the following is not the HCF of the numbers? **[1]**

   a) 75 b) 60

   c) 45 d) 90

5. The smallest number that has to be subtracted from 6143, so that it becomes exactly divisible by 10 is **[1]**

   a) 3 b) 2

   c) 4 d) 5

6. Which of the following is NOT a factor of 360? **[1]**

   a) 72 b) 45

   c) 80 d) 24

7. A number divisible by 3, 5, and 7 must be a multiple of: **[1]**

   a) 10 b) 34

   c) 105 d) 14

8. If HCF(72, x) = 12 and x < 72, which of the following is possible? **[1]**

   a) 18 b) 48

   c) 24 d) 30

9. What is the first common multiple of 3 and 5? **[1]**

   a) 12 b) 15

   c) 10 d) 20

10. LCM of 45 and 90 is **[1]**

a) 45 b) 15

c) 90 d) 180

11. If two numbers have HCF = 1 and LCM = 35, the numbers can be: **[1]**

a) 15, 21 b) 10, 14

c) 7, 14 d) 5, 7

12. Which of the following statements is not true? **[1]**

- The HCF of two distinct prime numbers is 1
- The HCF of two coprime numbers is 1
- The HCF of two consecutive even numbers is 2
- The HCF of an even and an odd number is even.

a) The HCF of two consecutive even numbers is 2

b) The HCF of two distinct prime numbers is 1

c) The HCF of two coprime numbers is 1

d) The HCF of an even and an odd number is even.

13. The largest cube that can fit exactly in a box of dimensions 18 cm × 24 cm × 30 cm has side: **[1]**

a) 6 cm b) 18 cm

c) 3 cm d) 12 cm

14. LCM of 5, 6 and 7 is **[1]**

a) 42 b) 30

c) 210 d) 35

15. LCM of 10, 15 and 20 is _____. **[1]**

a) 30 b) 60

c) 180 d) 90

16. If HCF(a, b) = a, then: **[1]**

a) a and b are co-prime b) a > b

c) a = b d) a divides b

17. What is the LCM of 20 and 28? **[1]**

a) 280 b) 70

c) 140 d) 560

18. The LCM of two numbers is 48. One number is 16. The other number can be: **[1]**

a) 8 b) 10

c) 20 d) 12

19. If one number divides the other exactly, then the LCM is: **[1]**

a) Smaller number b) Product of numbers

c) 1 d) Greater number

20. How many factors does the number 60 have? **[1]**

a) 16 b) 10

c) 12 d) 18

21. **Match the following** **[1]**

Column I	Column II
(i) 6	(a) Factor of 16
(ii) 21	(b) Prime number
(iii) 19	(c) Perfect number
(iv) 8	(d) Multiple of 7

a) (i)-(c), (ii)-(d), (iii)-(b), (iv)-(a) b) (i)-(a), (ii)-(c), (iii)-(d), (iv)-(b)

c) (i)-(c), (ii)-(b), (iii)-(a), (iv)-(d) d) (i)-(a), (ii)-(d), (iii)-(c), (iv)-(b)

22. **Match the following** **[1]**

Column I	Column II
(a) Lowest composite number	(i) 1
(b) Number of factors of a prime number	(ii) 2
(c) HCF of 15 and 27	(iii) 3
(d) HCF of 3 and 8	(iv) 4

a) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii) b) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

c) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i) d) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

23. **Assertion (A):** To minimise the number of bags, the weight of each bag should be equal to the HCF of the given quantities. **[1]**

Reason (R): Using the largest common factor gives fewer equal parts.

a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false. d) A is false but R is true.

24. **Assertion (A):** The shortest length of a toran made using strips of 6 cm and 8 cm is 24 cm. **[1]**

Reason (R): 24 is the greatest common factor of 6 and 8.

a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false. d) A is false but R is true.

25. **Assertion (A):** The largest square tile that can be used to cover a 12 ft × 16 ft room has side 4 ft. **[1]**

Reason (R): 4 is the smallest common factor of 12 and 16.

a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

26. **Assertion (A):** If one number is a multiple of the other, then the LCM of the two numbers is the larger number. **[1]**

Reason (R): The larger number already contains all the prime factors of the smaller number.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

27. **Assertion (A):** Kabamai will get free gajak again after 70 days. **[1]**

Reason (R): 70 is the lowest common multiple of 7 and 10.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

28. Why is the HCF of % and 275 equal to 1? **[1]**

29. Find the common factors of 24 and 36. **[1]**

30. Find the LCM of 4 and 10. **[1]**

31. Anshu weaves torans using 6-cm strips, Guna uses 8-cm strips. What is the first length where they match? **[1]**

32. List all the prime numbers between 10 and 20. **[1]**

33. How many square tiles of side 2 m are required to cover a rectangular floor of 6 m × 4 m? **[1]**

34. Find the HCF of 18 and 24. **[1]**

Section B

35. A number is divisible by both 5 and 12. By which another number will that number be always divisible? **[2]**

36. Find the HCF of 108 and 252. **[2]**

37. A number is divisible by 12. By what other number will that number be divisible? **[2]**

38. Find the HCF of 45 and 75 using prime factorisation. **[2]**

39. Is the first number divisible by the second? Use prime factorisation. **[2]**

a. 150 and 25

b. 84 and 12

40. A number has 8 and 14 as two of its factors. What other factors could that number have? **[2]**

41. Find the HCF of 70, 105, 175 **[2]**

42. You have 24 small bottles of lemonade and 36 large bottles. What is the largest number of equal-sized groups you can make if each group has the same number of small and large bottles? **[2]**

43. Find the LCM of 12, 16, 24 and 36? **[2]**

44. Find the LCM of 48 and 64. **[2]**

Section C

45. Find the H.C.F. of 144, 180 and 324. **[3]**

46. Find the H.C.F. of 28 and 72 by prime factorisation method. **[3]**

47. Who am I? I am a number less than **50**. One of my factors is **6**. The sum of my digits is 9. **[3]**

48. Which of the following numbers is the product of exactly three distinct prime numbers: 50, 84, 105, 280? **[3]**

49. Consider the numbers 72 and 144. Suppose they are factorised into composite numbers as: $72 = 6 \times 12$ and $144 = 8 \times 18$. Seeing this, can one say that these two numbers have no common factor other than 1? Why not? **[3]**

50. Find all the common factors and the HCF of 112 and 84. **[3]**

51. Find the H.C.F. of 14, 84 and 105. **[3]**

52. Find the LCM of 14 and 35 using prime factorisation. Show steps. **[3]**

53. Using division method, find the HCF of 300 and 150. **[3]**

54. Find the smallest number that is divisible by 6, 9, and 15. **[3]**

Section D

55. **Read the following text carefully and answer the questions that follow:** **[4]**

Three school bells ring at regular intervals. The first bell rings every 6 minutes, the second every 8 minutes, and the third every 12 minutes. At 9:00 a.m., all the bells ring together. Students want to find out after how many minutes all three bells will ring together again. The teacher explains that such problems can be solved using the concept of Least Common Multiple (LCM), which helps find the earliest time when events repeat together.

Questions:

a. What does LCM stand for? **(1)**

b. Why is LCM used in this problem? **(1)**

c. Find the LCM of 6, 8, and 12. **(2)**

OR

Why is HCF not suitable here? **(2)**

56. **Read the following text carefully and answer the questions that follow:** **[4]**

During a mathematics class, Ravi asked why prime numbers are important. Tanya explained that a prime number is greater than 1 and has only two factors. She gave examples like 7 and 11, which are prime numbers. She also explained that numbers such as 12 and 15 are not prime because they have more than two factors. The teacher added that every composite number can be written as a product of prime numbers, which makes prime numbers very important in mathematics.

Questions:

a. What is a prime number? **(1)**

b. Is 15 a prime number? Give reason. **(1)**

c. Explain why 7 is a prime number. **(2)**

OR

Write two composite numbers and justify. **(2)**

57. **Read the following text carefully and answer the questions that follow:** **[4]**

In a school library, there are 780 English books and 364 Science books. Ms. Yakang, the librarian, wants to arrange these books neatly on shelves so that each shelf contains the same number of books of each subject. She also wants to ensure that no books are left over. To do this, she needs to find a number that divides both 780 and 364 exactly. The students suggest using prime factorisation to find the highest common factor of the two numbers. This will help decide how many books can be placed on each shelf while maintaining equal distribution and proper organisation.

Questions:

a. What does HCF mean? **(1)**

b. Why is HCF suitable for this situation? **(1)**

c. Find the maximum number of books that can be placed on each shelf. **(2)**

OR

Explain why a number greater than the HCF cannot be used. **(2)**

Section E

58. A farmer has two fields of sizes 180 m and 252 m. He wants to divide both fields into smaller rectangular plots of the same size such that no land is wasted. What is the largest possible length of each plot? Also, how many plots will he get from each field? **[5]**

59. Find the common factors and the HCF of the following numbers: **[5]**

a. 36, 48

b. 84, 126

c. 45, 120

d. 132, 198

e. 121, 242

60. A rectangular hall measures 42 m by 30 m. You want to place square carpets of the largest possible size **[5]**

a. What should be the side of each carpet?

b. How many carpets will be needed?

61. Three bells ring at intervals of 12 minutes, 18 minutes, and 24 minutes. If they all ring together at 6:00 a.m., at what time will they ring together next? Find the answer using LCM and explain the steps. **[5]**

62. Make a general statement about the HCF for the following pairs of numbers. You could consider examples before coming up with general statements. Look for possible explanations of why they hold. **[5]**

a. Two consecutive even numbers

b. Two consecutive odd numbers

c. Two even numbers

d. Two consecutive numbers

e. Two co-prime numbers

Share your observations with the class.

63. Find the LCM of the following numbers: **[5]**

a. 24, 90

b. 40, 84

c. 66, 110, 165

d. 72, 120

e. 45, 60, 90

64. Mary has a dance class every 2nd day and painting class every 3rd day. On which of the day will she have both the classes? **[5]**

65. List all the factors of the following numbers: **[5]**

a. 72

b. 96

c. 120

d. 144

e. 210

66. Tick the correct statement(s). The LCM of two different prime numbers (m, n) can be: **[5]**

a. Less than both numbers

b. In between the two numbers

c. Greater than both numbers

d. Less than m $\times$ n

e. Greater than m $\times$ n

67. The LCM of 3 and 24 is 24 (it is one of the two given numbers). **[5]**

a. Find more such number pairs where the LCM is one of the two numbers.

b. Make a general statement about such numbers. Describe such number pairs using algebra.

Section F

68. **Fill in the blanks:** **[6]**

(a) The HCF and LCM of two numbers are related by the equation HCF $\times$ LCM = _____. **[1]**

(b) The numbers which are not multiples of 2 are known as _____. **[1]**

(c) Numbers that are neither prime nor composite are _____. **[1]**

(d) The number of factors of a prime number is _____. **[1]**

(e) Using prime factorization, the prime factors of 84 are 2, _____, and _____. **[1]**

(f) Using prime factorization, the LCM of 10 and 15 is _____. **[1]**